AI FOR GOOD MINOR | WEEK 7 HACKATHON

CigTracker
Cigarette-butt detection in natural areas

Finding visible cigarette litter in Dutch green spaces
with Roboflow and a local Streamlit app.

Pavitra Pradeep & Jalie Vermeegen

The problem

Cigarette litter affects green spaces. Staff need to decide where their time and budgets will have the most value & effect.

Small and scattered

Butts can blend into grass, leaves and soil. Counting them by hand across inspection photos takes time.

Plants and soil

Studies found reduced grass and clover growth, and nicotine moving from discarded butts into nearby plants.

Animals and fire

Ingestion can harm animals. Burning cigarettes can ignite dry vegetation. Wildlife evidence is limited.

Sources: Green et al. (2019), Selmar et al. (2018), Novotny et al. (2011), RCMP (2025)

Our user and their decision

A municipal maintenance coordinator

Reviews inspection photos of public green spaces and decides where cleaning teams should work.

Park rangers, landowners and volunteers can also use it.

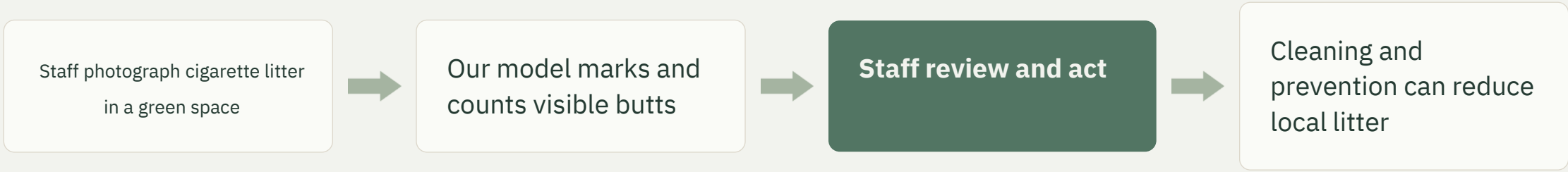
THEIR DECISION

Which location needs a cleanup, and where does recurring litter justify targeted prevention?

Limited camera access, computer access or setup skills can exclude users. Less-photographed areas can be overlooked.

SDG 15: Life on Land

Target 15.5: reduce habitat degradation and halt biodiversity loss.



Where it applies

Forests, parks, gardens and small urban green spaces in the Netherlands.

Why detection helps

Repeated inspections can reveal places that need attention. Staff record locations and dates separately.

Our contribution

Indirect habitat protection. We do not measure pollution levels or prove biodiversity improves.

Source: United Nations, SDG 15, target 15.5

What we built

1

Upload

The user selects one photo in Streamlit.

2

Detect

The local RF-DETR model predicts cigarette-butt boxes and scores.

3

Filter

A confidence threshold removes lower-scoring predictions.

4

Review

The app shows bounding boxes and a count of possible butts.

5

Download

The user downloads the annotated image.

Roboflow performs the training. The exported model runs locally on the CPU.

`One photo at a time. No API key or Roboflow inference credits required.`

Repeated photo review is the intended use. Batch processing and automatic mapping are not built in.

Our dataset and the app

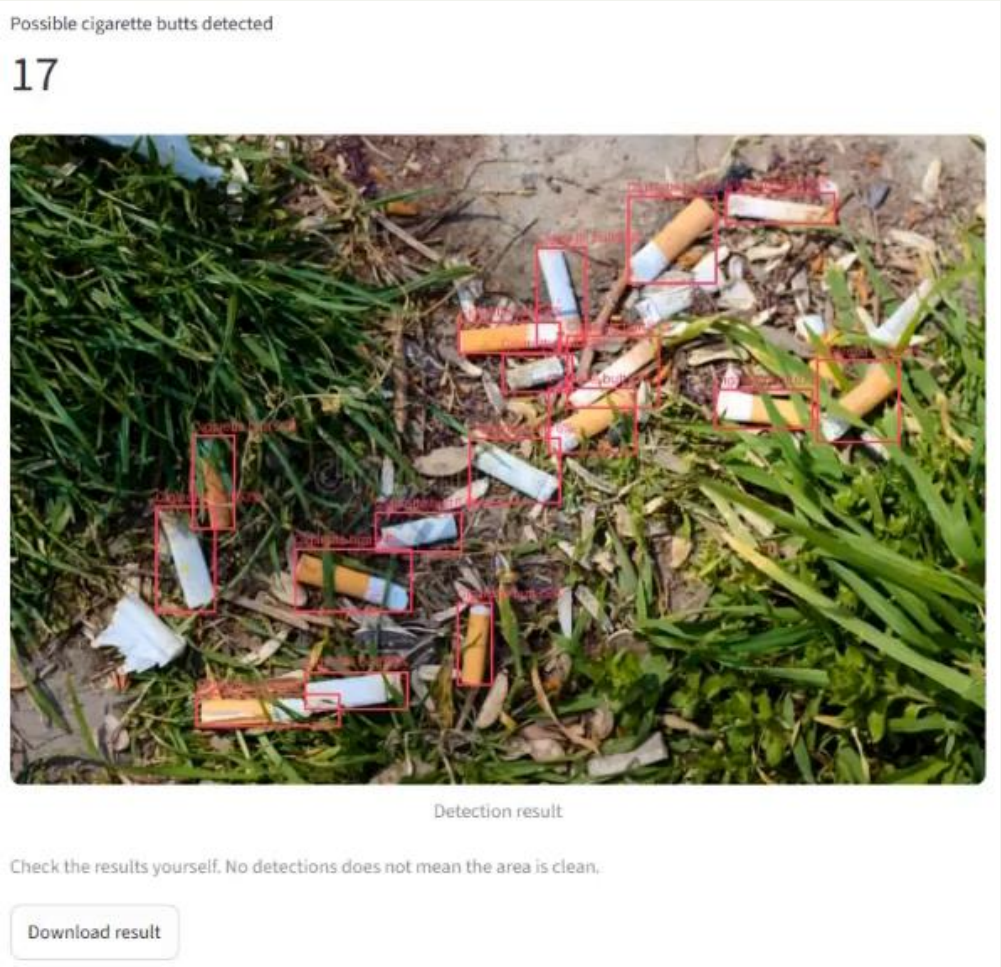
320 images we labelled ourselves

- 224 training images
- 64 validation images
- 32 test images

One class: cigarette_butts

Own photos and video from public spaces

Keep related video frames in one split



What does 47% confidence mean?

Minimum score

For Medium, we show predictions with a model score of at least 0.47. Lower-scoring boxes stay hidden.

The tradeoff

Raising the threshold usually reduces false alarms, but can also hide more real cigarette butts.

What the score means

The model score is not a verified probability, and 47% is not the model’s accuracy.

Roboflow suggested 37% for Small and 47% for Medium based on F1. Tune thresholds on validation data.

A larger independent test set improves evaluation reliability, not model accuracy by itself.

Testing: RF-DETR Small and Medium

Reported test-set results at each model's suggested confidence threshold.

Model	Precision	Recall	F1	mAP@50	Threshold
Small	80.2%	62.0%	69.9%	69.8%	37%
Medium	**90.1%**	**60.7%**	**72.5%**	**71.8%**	**47%**

Medium found 91 butts versus Small's 93, with 10 false positives versus 23.

We chose Medium: nearly the same recall, fewer false alarms, and higher F1 and mAP@50.

The error that matters most

A missed butt can make a polluted location appear cleaner than it really is.

False negatives: 59

False positives: 10

Why recall matters

Medium found 60.7% of labelled butts. Missing litter can cause an area to receive too little attention.

Why precision still matters

Medium’s 90.1% precision reduces false alarms that could waste staff time or trigger an unnecessary cleanup.

Limits and ethics

Limited conditions

No clean-scene images. Small or obscured butts and unfamiliar backgrounds may cause errors. No photo’s in dataset of different seasons, times, etc.

Personal details

We removed as much identifiable detail as possible while labelling. Further privacy checks remain advisable.

Already implemented

Local processing, adjustable threshold, and a warning that no detections does not mean clean.

Recommended next steps

Add varied and clean-scene images. Verify video-frame separation. Test independent new photos and human review.

Staff should check results before acting. Record locations separately and combine detections with regular inspections.

Detection supports inspections. People decide where to act.

Live demo: upload, detect, review and download.

Pavitra Pradeep & Jalie Vermeegen · Code, weights and demo linked in the README